

Open LLM Evaluation Framework - Final Report

Introduction

The Open LLM Evaluation Framework is a research-oriented project designed to evaluate the performance and reasoning capability of open-source Large Language Models (LLMs) using standardized benchmarks and evaluation metrics.

Research Question

Does instruction tuning impact reasoning accuracy in open-source LLMs of varying sizes?

Objectives

- Evaluate mathematical reasoning capabilities of open-source LLMs
- Compare base models vs instruction-tuned models
- Measure accuracy, consistency, and reliability
- Analyze failure patterns through error analysis
- Build a reproducible evaluation pipeline

Models Evaluated

Model	Parameters	Type
DistilGPT2	82M	Base Model
GPT2-Medium	345M	Base Model
TinyLlama-1.1B-Chat	1.1B	Instruction Tuned

Methodology

- 1. Dataset loading — GSM8K 10 sample questions
- 2. Model loading — HuggingFace Transformers
- 3. Answer generation — each model answers each question
- 4. Number extraction — regex based extraction from output
- 5. Accuracy calculation — extracted vs expected answer
- 6. Consistency check — same results across multiple runs
- 7. Error analysis — failure pattern identification
- 8. Visualization — charts and heatmaps generated
- 9. Report generation — CSV and markdown reports

Benchmark Used

GSM8K (Grade School Math 8K)

Mathematical reasoning benchmark with arithmetic word problems. Tests model ability to understand questions and compute correct numerical answers.

Evaluation Metrics

Metric Description
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Accuracy Correct answers / Total questions
Consistency Same output across multiple runs
Reliability Overall dependability of responses
Hallucination Rate Wrong/unsupported outputs

Experimental Results

GSM8K Accuracy

Model	Parameters	Type	Correct	Total	Accuracy
DistilGPT2	82M	Base	0	10	0%
GPT2-Medium	345M	Base	0	10	0%
TinyLlama-1.1B	1.1B	Instruction Tuned	3	10	30%

Consistency Results

Model	Run 1	Run 2	Run 3	Consistent
DistilGPT2	0%	0%	0%	Yes
GPT2-Medium	0%	0%	0%	Yes
TinyLlama	30%	30%	30%	Yes

Error Analysis Summary

Model	Primary Failure Pattern
DistilGPT2	Extracts numbers from question instead of computing
GPT2-Medium	Same pattern — no mathematical reasoning ability
TinyLlama	Correct on simple addition, fails on subtraction/multiplication

Key Findings

Finding 1 — Instruction Tuning is Critical

Base models (DistilGPT2, GPT2-Medium) scored 0% regardless of size. Only instruction-tuned TinyLlama showed 30% accuracy.

Finding 2 — Model Size Alone is Not Enough

GPT2-Medium (345M) = same as DistilGPT2 (82M) = 0%.

4x larger model showed zero improvement without instruction tuning.

Finding 3 — Operation Type Matters

TinyLlama succeeded on simple addition but failed on subtraction, multiplication, and multi-step problems.

Generated Outputs

- `results/report/gsm8k\_results\_summary.csv`
- `results/report/accuracy\_chart.png`
- `results/report/model\_comparison.png`
- `results/report/hallucination\_heatmap.png`
- `results/report/error\_analysis.md`

Observations

- Instruction tuning is more important than model size
- Base models have zero mathematical reasoning ability
- TinyLlama shows promising results for a 1.1B model
- Consistent results across multiple evaluation runs
- Modular framework supports easy benchmark expansion

Future Work

- Add Mistral-7B evaluation (instruction tuned, 7B)
- Integrate TruthfulQA benchmark
- Integrate HellaSwag benchmark
- GPU support for larger models
- Increase sample size to 100+ questions
- Build interactive dashboard

Conclusion

This study demonstrates that instruction tuning is the most critical factor for mathematical reasoning in open-source LLMs — more important than model size alone.

TinyLlama-1.1B-Chat achieved 30% accuracy on GSM8K while base models (DistilGPT2 82M, GPT2-Medium 345M) scored 0%, clearly answering our research question:

- > Instruction tuning significantly impacts reasoning accuracy,
- > regardless of model size.

References

- [GSM8K Paper](<https://arxiv.org/abs/2110.14168>)
- [TruthfulQA Paper](<https://arxiv.org/abs/2109.07958>)
- [HuggingFace Transformers](<https://github.com/huggingface/transformers>)
- [TinyLlama](<https://github.com/jzhang38/TinyLlama>)

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